Anwoy Barua

Chittagong, Bangladesh
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EDUCATION

Master of Science in Statistics July 2025- June 2026
*University of Chittagong| CGPA: **3.67/4** (3rd)*

Bachelor of Science in Statistics Jan. 2020 – May 2025
*University of Chittagong| CGPA: **3.63/4** (5th)*

RESEARCH INTERESTS

- Climate extremes, climate risk assessment, and impact modeling
- Climate model evaluation, bias correction, and uncertainty quantification
- Climate change impacts on public health and climate-sensitive health risk assessment

PUBLICATIONS

1. **Barua, A.**, Ghosh, S., and Karmakar S. **''Evaluation of Statistical and Machine Learning-Based Bias Correction Techniques for CMIP6 Daily Air Temperature over Bangladesh.''** *Manuscript under preparation.*

   LightGBM demonstrated superior overall performance and improved correction of spatial and extreme temperature compared to Quantile Mapping methods.

2. **Barua, A.**, Ghosh, S., and Karmakar S. **''Validation of ERA5 and ERA5-Land Reanalysis Data for Daily Air Temperature and Extreme Indices over Bangladesh.''** *Under Review at Theoretical and Applied Climatology.* View Preprint

   ERA5 generally provided better representation of daily temperature and extreme magnitudes, whereas ERA5-Land showed comparable performance for long-term percentile-based temperature extremes.

Research Experience

Research Assistant (Part-time) Sep. 2026 – Present
Working under the supervision of Associate Professor Sanjib Ghosh.

TECHNICAL SKILLS

- **Programming:** Python (xarray, pandas, numpy, matplotlib, seaborn, scikit-learn), and R (ggplot2, dplyr)
- **Tools:** ArcGIS (ArcMap10.5), Google Earth Pro, Microsoft Excel, SPSS
- **Data Skills:** Statistical analysis and visualization, Machine Learning, Geospatial analysis, Remote sensing fundamentals
- **Database:** PostgreSQL

LINGUISTIC PROFICIENCY

Bengali: Native
IELTS Academic: Overall **7.5** (Listening - 8.5, Reading - 7.5, Writing - 7, Speaking - 6.5) Certificate Link

AWARDS

Information Technology Fellowship for Master's Thesis Apr. 2026- Sep. 2026
*Awarded by **ICT Division**, Government of Bangladesh*

PROFESSIONAL TRAINING

Mastering GIS with ArcGIS: from Fundamentals to Real-World Projects Aug. 2025 - Dec. 2025
by Data Solution 360 Certificate Link

- Hands-on training in spatial analysis, geoprocessing, and remote sensing fundamentals
- Link to some of the maps created during this bootcamp. Project link

Digital Diversity, Equity, and Inclusion (DDEI) Mentorship Programme Oct. 2023- Aug. 2024
by BRAC James P Grant School of Public Health (JPSGH) Certificate Link

- Gained practical experience in literature review, research question formulation, and quantitative research design.

LEADERSHIP EXPERIENCE

President Feb. 2024- Apr. 2025
Programming Club Linkedin Page
Department of Statistics, University of Chittagong

- Organized Python and R workshops
- Mentored students in programming and data analysis

CERTIFICATIONS

- MOOC on "**Spatial Data Science: The New Frontier in Analytics**" — Esri, 2025 Certificate Link
- Short course on "**Basic Quantitative Research methodology**" — BRAC JPGSPH, 2024 Certificate Link
- **Introduction to Systematic Review and Meta Analysis in Health Research** — BRAC JPGSPH, 2024 Certificate Link

REFERENCES

1. **Sanjib Ghosh** *(Thesis Supervisor)*
   Associate Professor
   Department of Statistics, University of Chittagong
   E-mail: sanjib.stat@cu.ac.bd

2. **Mrittika Barua, PhD**
   Adjunct Assistant Professor and Technical Advisor
   Centre for Science of Implementation and Scale-up
   BRAC James P. Grant School of Public Health, BRAC University
   E-mail: mrittika@bracu.ac.bd